Audio Cheat Sheet: Yamaha CL5

Sound check, gain, HPF and EQ. Starting points, not presets. Trust your ears.

Sound check

1. **Start muted.** Unmute only the channel you are checking, +48V only for condensers. Have fader all the way down to -∞.
2. **[SEL] the channel** and have the player play at full performance level.
3. **Set GAIN** so the -18 dB LED lights on peaks, -12 dB rarely. Average signal at -20 dB.
4. **Phase [Ø]** only if two mics on one source sound thin (kick, snare). See if turing *invert phase* on makes it not sound muddy.
5. **HPF on** in Selected Channel view. Start with the table value below.
6. **EQ** in small moves. Mainly remove/cut bad sounding frequencies.
7. **Fader up,** and balance against the full band. Listen to each souce in context with other audio sources. (Vocals are loud and center)
8. **Check overall volume,** keep all faders around Unity (0 dB) for the most fader percision. Adjust the master output for volume of the overall room volume.

Gain Staging

- **OVER** — NEVER EVER!
- **-6** — Never
- **-12** — Rarely
- **-18** — Occasionally
- **-30** — Always (or almost always)
- **-60** — Always (when there's signal)

Channel meter LEDs: -18 lights occasionally on peaks, -12 rarely, -6 never. OVER never.

Set gain once. Use the fader during the service, not gain.

Electronic Instruments: set their volume to about 70-80%, leaving room to turn up.(Keyboards, Electric Guitars, etc.)

[CUE] lets you listen in headphones without sending to the house.

What each range does

20–100 Hz	100–300 Hz	300–800 Hz	0.8–2 kHz	2–5 kHz	5–8 kHz	8–16 kHz
Rumble, sub weight	Body, warmth, boom	Mud, boxiness	Honk, nasal tone	Presence, attack, harsh	Sibilance, bite	Air, shimmer

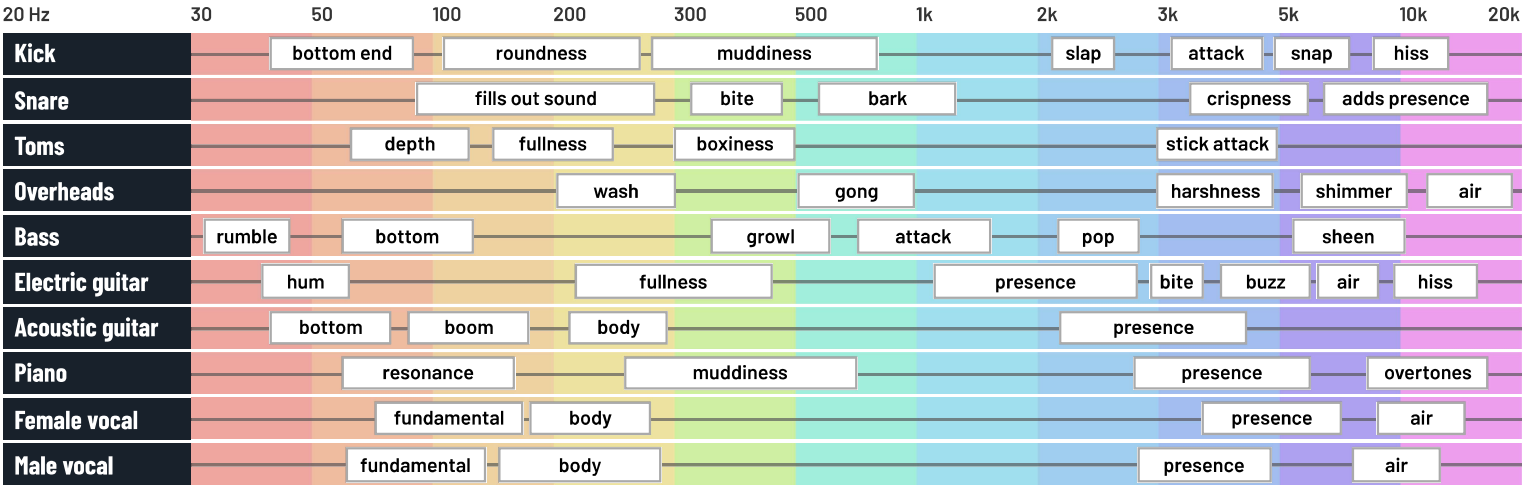
HPF and EQ by input

Input	HPF	Body / weight	Clean up if needed	Clarity / detail
Kick	**35 Hz**	50–80 Hz thump	200–400 Hz boxy	2–5 kHz beater click
Snare	**90 Hz**	100–200 Hz fatness	400–600 Hz boxy	4–6 kHz crack, 8–10 kHz snap
Hi-hat	**300 Hz**	Use HPF to reduce bleed	3–5 kHz clank	8–12 kHz shimmer
Toms	**70 rack / 50 floor**	80–250 Hz fullness	300–500 Hz box	3–5 kHz stick attack
Overheads	**150 Hz**	Use HPF	200–300 Hz wash, 3–5 kHz harsh	10 kHz+ air
Bass	**30 Hz**	40–80 Hz weight, 100–250 Hz warmth	250–400 Hz mud	700 Hz–1 kHz definition, 2–3 kHz clank
Electric guitar	**90 Hz**	100–200 Hz body	200–300 Hz mud, 500–800 Hz honk, 5 kHz+ fizz	2–4 kHz bite (dip to make room for vocals)
Acoustic guitar	**90 Hz**	100–250 Hz warmth	100–150 Hz boom, 200–300 Hz mud	2–5 kHz pick, 8–12 kHz sparkle
Piano	**50 Hz**	60–100 Hz weight	250–400 Hz mud	2–4 kHz attack, 8 kHz+ air
Vocals	**120 male / 150 female**	150–250 Hz warmth	200–400 Hz mud, 500 Hz–1 kHz nasal, 5–8 kHz sibilance	2–4 kHz clarity, 10 kHz+ air
Speaking mic	**120 Hz**	Use HPF	100–150 Hz boom, 200–300 Hz mud, 5–8 kHz sibilance	2–4 kHz clarity

HPF: raise it until the unwanted lows clear, then back off if the source loses body. Toggle it on and off to check.

Frequency map: where each part lives

Columns are equal width, not to scale. When two parts crowd the same range, cut a little from the less important one.



When in doubt, ask the lead engineer before changing the main mix.

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